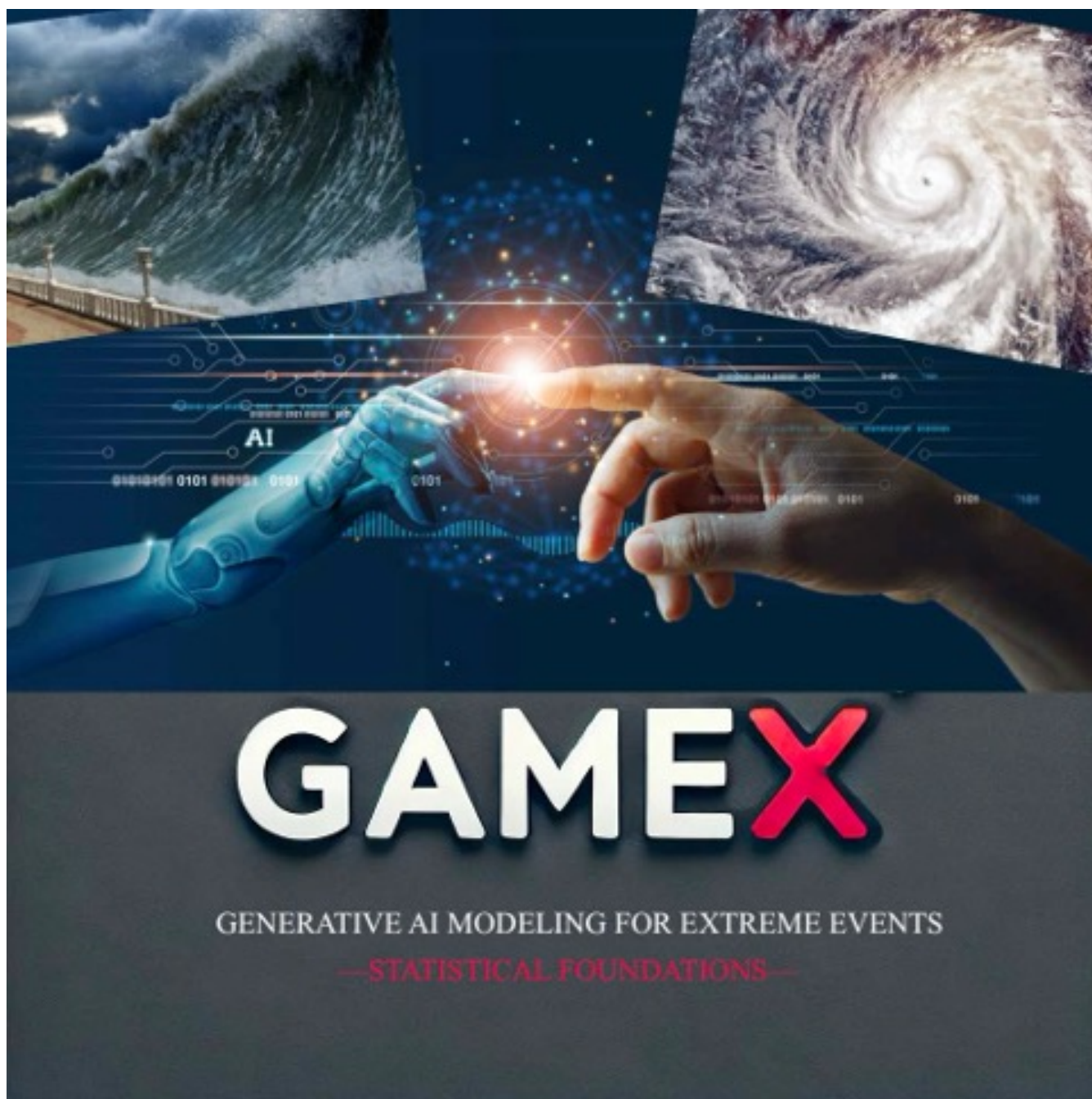


Edinburgh Summer School on Generative AI for Extremes



THE UNIVERSITY
of EDINBURGH



Programme—updated 6 September 2026
8–11 September 2026

<https://gamex-network.github.io/school/>

GL²EN (Glasgow–Lancaster–London–Edinburgh–Newcastle) Extremes Network

GAMEX Project Members: Luca Trapin • Miguel de Carvalho • Massimo Ventrucci • Ivo Bonfanti • Johnny Lee • Giorgia Ramponi
Till Freihaut • Brendan Murphy • Luis Gutierrez Inostroza • Clemente Ferrer

Chair

Professor Miguel de Carvalho

Chair of Statistics & Data Science

Elected Fellow of Generative AI Laboratory

Organisers

Johnny Lee (Edinburgh), Lambert De Monte (Edinburgh, KAUST), Jordan Richards (Edinburgh), Daniela Castro-Camilo (Glasgow), Mengran Li (Glasgow), Teng Wei Yeo (Glasgow), and Luca Trapin (Bologna).

Welcome

Building on the success of GAME 2025 (<https://edin.ac/4akncn7>) in Edinburgh, the first worldwide workshop on generative AI modelling for extreme events, this summer school is led by GL²EN (Glasgow–Lancaster–London–Edinburgh–Newcastle) Extremes Network and brings PhD students and early-career researchers to the interface of generative AI and extreme value theory. It is an official ECAS course (European Courses in Advanced Statistics).

Partners



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Further Reading

In addition to the materials made available by the instructors, we recommend selected parts of the *Handbook of Statistics of Extremes* (<https://extremestats.github.io/>) and parts of the recent special issue on *Bridging Heavy Tails and AI* (<https://edin.ac/4akncn7>).

Day 1: Tuesday

8 September

Theme: *Foundations and Preparations for Generative Modelling*

Moderator: Luca Trapin

Time	Session	Lead
09:00–09:25	Registration & Welcome Coffee	Organisers
09:25–09:30	<i>Welcome & Overview</i>	Chair
09:30–11:00	<i>Introducing Extreme Value Theory</i>	Miguel de Carvalho
11:00–11:30	Coffee break	
11:30–12:30	<i>On Neural Generative Models</i>	Miguel de Carvalho
12:30–14:00	Lunch	
14:00–15:00	<i>Probabilistic Aspects of Diffusions</i>	Sotirios Sabanis
15:00–15:30	Coffee break	
15:30–17:00	<i>Practical: Neural Generative Models in Practice</i>	Johnny Lee

Day 2: Wednesday

9 September

Theme: *Generative Modelling for Extremes and Risk*

Moderator: Miguel de Carvalho

Time	Session	Lead
09:30–10:30	<i>GAN-based Generative Approaches for Extremes</i>	Stéphane Girard (online)
10:30–11:00	Coffee break	
11:00–12:00	<i>GAN-based Modelling for Operational Risk</i>	Luca Trapin
12:00–13:30	Lunch	
13:30–15:30	<i>Practical: Generative Modelling for Extremes and Risk</i>	Jordan Richards
15:30–16:00	Coffee break	
16:00–16:10	<i>Group photo</i>	
16:10–17:10	<i>Open Practice & Office Hours</i>	Instructors

Day 3: Thursday

10 September

Theme: *Multivariate Extremes, Cascades, and Simulation*

Moderator: Giorgia Ramponi

Time	Session	Lead
09:30–10:30	<i>Cascades, Multivariate Extremes, and Transformers</i>	Miguel de Carvalho
10:30–11:00	Coffee break	
11:00–12:00	<i>Multivariate Angular Simulation</i>	Emma Simpson
12:00–13:30	Lunch	
13:30–15:30	<i>Practical: Transformer-Based Approaches for Extremes</i>	Clemente Ferrer
15:30–16:00	Coffee break	
16:00–17:00	<i>Open Practice & Office Hours</i>	Instructors

Day 4: Friday

11 September

Theme: *Other Frontier Topics*

Moderator: Emma Simpson

Time	Session	Lead
09:30–10:30	<i>Reinforcement Learning</i>	Giorgia Ramponi
10:30–11:00	Coffee break	
11:00–12:00	<i>Pre-trained Extreme Value Modelling</i>	Jordan Richards
12:00–12:10	<i>Final Announcements</i>	—
12:10–12:20	<i>Closing Remarks</i>	Chair
12:20–13:50	Lunch	

[†] Lectures: Lecture Theatre B; practical sessions: Teaching Studio 3217.